

Problema 2:

Para las siguientes CFGs realice lo solicitado, mostrando todo el procedimiento.

1. $S \rightarrow 0A0 \mid 1B1 \mid BB$

$A \rightarrow C$

$B \rightarrow s \mid A$

$C \rightarrow s \mid \varepsilon$

a) elimine las producciones- ε

$S \rightarrow 0A0 \mid 1B1 \mid BB$

$A \rightarrow \varepsilon$

$B \rightarrow s \mid A$

$C \rightarrow s$

$S \rightarrow 00 \mid 1B1 \mid BB$

$A \rightarrow C$

$B \rightarrow s \mid \varepsilon$

$C \rightarrow s$

$S \rightarrow 0A0 \mid 00 \mid 11 \mid B \mid \varepsilon$

$A \rightarrow C$

$B \rightarrow s \mid A$

$C \rightarrow s$

$S \rightarrow 0A0 \mid 00 \mid 1B1 \mid 11 \mid B \mid BB$

$A \rightarrow C$

$B \rightarrow \varepsilon \mid A$

$C \rightarrow \varepsilon$

$S \rightarrow 0A0 \mid 00 \mid 1B1 \mid 11 \mid B \mid BB$

$A \rightarrow C$

$B \rightarrow s \mid A$

$C \rightarrow s$

b) elimine cualquier producción unitaria de la gramática resultante.

(A, A)

(C, B)

(C, C)

(A, A)

$A \rightarrow C$

(A, C)

(A, C)

$C \rightarrow s$

(A, s)

(B, B)

$B \rightarrow s$

(B, s)

(B, B)

$B \rightarrow A$

(B, A)

(B, A)

$A \rightarrow C$

(B, C)

(C, C)

$C \rightarrow s$

(C, s)

(A, A) $A \rightarrow C$

(A, C) $C \rightarrow s$

(A, s) $S \rightarrow 0A0 \mid 00 \mid 1B1 \mid 11 \mid B \mid BB$ $A \rightarrow 0A0 \mid 00 \mid 1B1 \mid 11 \mid B \mid BB$

(B, B) $B \rightarrow s \mid A$

(B, A) $A \rightarrow C$

(B, s) $S \rightarrow 0A0 \mid 00 \mid 1B1 \mid 11 \mid B \mid BB$ $B \rightarrow 0A0 \mid 00 \mid 1B1 \mid 11 \mid B \mid BB$

(C, C) $C \rightarrow s$

(C, s) $C \rightarrow 0A0 \mid 00 \mid 1B1 \mid 11 \mid B \mid BB$ $C \rightarrow 0A0 \mid 00 \mid 1B1 \mid 11 \mid B \mid BB$

$S \rightarrow 0A0|00|1B1|11|BB$
 $A \rightarrow 0A0|00|1B1|11|BB$
 $B \rightarrow 0A0|00|1B1|11|BB$
 $C \rightarrow 0A0|00|1B1|11|BB$

c) elimine cualquier símbolo inútil de la gramática resultante.

a. Remueva símbolos que no producen

$$T = \{0, 1\}$$

$$W_1 = \{S, A, B, C\}$$

b. Remueva símbolos no alcanzables.

$$Y_1 = \{S\}$$

$$Y_2 = \{A, B, 0, 1, S\}$$

$$Y_3 = \{A, B, 0, 1, S\}$$

$S \rightarrow 0A0|00|1B1|11|BB$
 $A \rightarrow 0A0|00|1B1|11|BB$
 $B \rightarrow 0A0|00|1B1|11|BB$

d) Coloque la CFG resultante en la Forma Normal de Chomsky

$$Z = 1$$

$$Y = 0$$

$S \rightarrow YAY|YY|ZBZ|ZZ|BB$
 $A \rightarrow YAY|YY|ZBZ|ZZ|BB$
 $B \rightarrow YAY|YY|ZBZ|ZZ|BB$

$$L_1 \rightarrow YA$$

$$L_2 \rightarrow ZB$$

$S \rightarrow L_1Y|YY|L_2Z|ZZ|BB$
 $A \rightarrow L_1Y|YY|L_2Z|ZZ|BB$
 $B \rightarrow L_1Y|YY|L_2Z|ZZ|BB$
 $L_1 \rightarrow YA$

$$L_2 \rightarrow ZB$$

$$Z = 1$$

$$Y = 0$$

R

2. $S \rightarrow aAa \mid bBb \mid \varepsilon$
 $A \rightarrow C \mid a$
 $B \rightarrow C \mid b$
 $C \rightarrow CDE \mid \varepsilon$
 $D \rightarrow A \mid B \mid ab$

a) elimine las producciones- ε

$S \rightarrow aAa \mid bBb \mid \varepsilon$	$S \rightarrow aa \mid bBb \mid \varepsilon$	$S \rightarrow aa \mid aAa \mid bb \mid bBb$	$S \rightarrow aAa \mid aa \mid bBb \mid bb$
$A \rightarrow \varepsilon \mid a$	$A \rightarrow C \mid a$	$A \rightarrow C \mid a$	$A \rightarrow C \mid a$
$B \rightarrow \varepsilon \mid b$	$B \rightarrow \varepsilon \mid b$	$B \rightarrow C \mid b$	$B \rightarrow C \mid b$
$C \rightarrow DE$	$C \rightarrow CDE \mid DE$	$C \rightarrow CE \mid E$	$C \rightarrow CDE \mid CE \mid DE \mid E$
$D \rightarrow A \mid B \mid ab$	$D \rightarrow \varepsilon \mid B \mid ab$	$D \rightarrow A \mid B \mid ab$	$D \rightarrow A \mid B \mid ab$

b) elimine cualquier producción unitaria de la gramática resultante.

$(S, S) (A, A) (B, B) (C, C) (D, D)$

(A, C)	$C \rightarrow CDE \mid CE \mid DE \mid E$	$A \rightarrow CDE \mid CE \mid DE \mid a$
(B, C)	$C \rightarrow CDE \mid CE \mid DE \mid E$	$B \rightarrow CDE \mid CE \mid DE \mid b$
(D, A)	$A \rightarrow CDE \mid CE \mid DE \mid a$	$D \rightarrow CDE \mid CE \mid DE \mid a \mid ab$
(D, B)	$B \rightarrow CDE \mid CE \mid DE \mid b$	$D \rightarrow CDE \mid CE \mid DE \mid a \mid b \mid ab$

$S \rightarrow aAa \mid aa \mid bBb \mid bb$
 $A \rightarrow CDE \mid CE \mid DE \mid a$
 $B \rightarrow CDE \mid CE \mid DE \mid b$
 $C \rightarrow CDE \mid CE \mid DE$
 $D \rightarrow CDE \mid CE \mid DE \mid a \mid b \mid ab$

c) elimine cualquier símbolo inútil de la gramática resultante

a. Remueva símbolos que no producen

$T = \{a, b\}$

$W_1 = \{A, B, D, S\}$

$C \rightarrow$ no produce

$E \rightarrow$ no produce

$S \rightarrow aAa \mid aa \mid bBb \mid bb$

$A \rightarrow a$

$B \rightarrow b$

$D \rightarrow a \mid b \mid ab$

b. Remueva símbolos no alcanzables.

$$V_1 = \{S\}$$

$$V_2 = \{a, b, A, B, S\}$$

$0 \rightarrow$ no es derivable

$$S \rightarrow aAa|aa|bBb|bb$$

$$A \rightarrow a$$

$$B \rightarrow b$$

d) Coloque la CF6 resultante en la Forma Normal de Chomsky

$$S \rightarrow aAa|aa|bBb|bb$$

$$A \rightarrow a$$

$$B \rightarrow b$$

$$S \rightarrow AAA|AA|BBB|BB$$

$$A \rightarrow a$$

$$B \rightarrow b$$

$$G_1 \rightarrow AA$$

$$G_2 \rightarrow BB$$

$$S \rightarrow G_1A|G_1|G_2B|G_2$$

$$A \rightarrow a$$

$$B \rightarrow b$$

$$G_1 \rightarrow AA$$

$$G_2 \rightarrow BB$$

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3.

Actividad: Calcular forma normal

$$S \rightarrow ASA|aB$$

$$A \rightarrow B|S$$

$$B \rightarrow b|a$$

1. Eliminar epsilon

$$S \rightarrow ASA|aB$$

$$A \rightarrow \epsilon|S$$

$$S \rightarrow S|aB$$

$$S \rightarrow ASA|S|aB|a|AS|SA$$

$$A \rightarrow B|S$$

$$B \rightarrow b$$

2. Uniques

$$(S, S) \leftarrow \text{no es}$$

$$(A, A)$$

$$(B, B) \leftarrow \text{no es}$$

$$(A, A) \quad A \rightarrow B$$

$$(A, A) \quad A \rightarrow S$$

$$(A, B)$$

$$(A, S)$$

$$(A, B) \quad B \rightarrow b$$

$$(A, S) \quad S \rightarrow ASA|S|aB|a$$

$$A \rightarrow b$$

$$A \rightarrow \epsilon|SA$$

$$S|A|SA|aB|a$$

$$S \rightarrow ASA|S|AS|SA|aB|a$$

$$A \rightarrow b|ASA|S|AS|SA|aB|a$$

$$B \rightarrow b$$

3. Inútiles e Irreducibles

$$T = \{a, b\}$$

$$W_1 = \{S, A, B\} \text{ no hay inútiles}$$

$$V_1 = \{S\}$$

$$V_2 = \{a, b, A, B, S\} \text{ no hay inútiles}$$

$$S \rightarrow ASA | S | AS | SA | aB | a$$

$$A \rightarrow b | ASA | S | AS | SA | aB | a$$

$$B \rightarrow b$$

4. Chomsky

$$Z \rightarrow b$$

$$Y \rightarrow a$$

$$S \rightarrow ASA | S | AS | SA | YB | Y$$

$$A \rightarrow Z | ASA | S | AS | SA | YB | Y$$

$$B \rightarrow Z$$

$$C_1 \rightarrow SA$$

$$S \rightarrow AC_1 | S | AS | SA | YB | Y$$

$$A \rightarrow Z | AC_1 | S | AS | SA | YB | Y$$

$$B \rightarrow Z$$